Yusuke Izawa, Assistant Professor in Computer Science

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Research Interest: Programming Language and Software

- interpreters, compilers, just-in-time (JIT) compilation
- meta-compilation, the RPython/PyPy and Truffle/GraalVM ecosystems
- Live programming environment for cyber-physical devices

Education

2020-2023 **Ph.D. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *Supporting multi-scope and multi-level compilation in a meta-tracing just-in-time compiler.* (GPA 3.49)

2018-2020 **M.Sc. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *Stack Hybridization: A Mechanism for Bridging Two Compilation Strategies in a Meta Compiler Framework.*

2014-2018 **B.Sc. Mathematical and Computing Science, Tokyo Institute of Technology**.
Thesis title: *BacCaml: A Meta-JIT Compiler Based on Both Tracing and Method JIT Compilations.*

Employment History

- 2025.06 – 2025.09 Heinrich-Heine-Universität Düsseldorf, Guest Researcher.
- 2024.04 – now Tokyo Metropolitan University, Assistant Professor.
- 2023.04 – 2024.03 IBM Research – Tokyo, Research Scientist.
- 2023.04 – 2023.04 JSPS Research Fellow PD. (declined)
- 2021.08 – 2021.10 IBM Research – Tokyo, Research Internship (Paid).
- 2021.04 – 2023.03 JSPS Research Fellow DC2.
- 2020.11 – 2023.03 Tokyo Institute of Technology, Dept. of Math. and Comp., Research Assistant.

Grants, Honours and Scholarships

2025
- **Shortlisted Candidate**, Tenure-Track Assistant Professor in Computer Engineering, Aalto University (shortlisted from 55 applicants).
- **Tokyo Metropolitan University, Funds for Research Stay in Overseas**. Living expenses are covered by Tokyo Metropolitan University Fund.
- **Graint-in-Aid for Early-Career Scientists**. Research expenses are converd by KAKENHI.

2024
- **Graint-in-Aid for Research Activity Start-up**. Research expenses are converd by KAKENHI.

2023
- **Long-Term Performance Award.** Award from IBM, based on skills needed for the future, long-term caree potential, and strong indivisual performance.
- **Research Fellowship for Young Scientists (JSPS PD).** Fellowship from the Japan Society for the Promotion of Science (JSPS), covering living expenses. Research expenses covered by KAKENHI. (declined)

2021
- **Research Fellowship for Young Scientists (JSPS DC2)**. Fellowship from the Japan Society for the Promotion of Science (JSPS), covering living expenses. Research expenses covered by KAKENHI.

2020
- **JST Strategic Basic Research Programs ACT-X**. Research expenses covered by Japan Science and Technology Agency (JST).

Grants, Honours and Scholarships (continued)

2019 **2nd Place, Graduate Category, ACM Student Research Competition,** ***Association for Computing Machinery*****. [*]**

Academic Services

- 2026 Program Committee, PEPM 2026.
- 2025 – now Steering Committee, VMIL.
- 2025 – nw Editor, Application Working Group on IPSJ Journal (Japanese).
- 2025 Organizer and Program Co-Chair, VMIL 2025.
- Program Committee, Programming and Programming Language Workshop (PPL, Japanese).
- 2024 External Reviewer, The Programming Journal, Volume 8. Issue 2.
- Reviewer, ACM Transactions on Architecture and Code Optimization.
- Reviewer, ACM Transactions on Software Engineering and Methodology.
- Program Committee, MPLR 2024, ICCQ 2024, MoreVMs 2024.
- 2023 Program Committee, ICSME 2023 (Industry Track), ICCQ 2023.
- Artifact Evaluation Committee, The Programming Journal, Volume 8.
- 2022 Artifact Evaluation Committee, The Programming Journal, Volume 7.
- 2021 Artifact Evaluation Committee, PACT 2021, ECOOP 2021.
- 2020 Member of Student Volunteer, SPLASH 2020.
- Co-reviewer of Onward! Essays, SPLASH 2020.
- External reviewer of Scala Symposium, ECOOP 2020.
- Candidate of Programming Language Mentoring Workshop, PLDI 2020.
- 2019 Member of Student Volunteer, Programming 2019.

Publications (Refereed)

Journal

1. Yusuke Izawa, Hidehiko Masuhara, and Carl Friedrich Bolz-Tereick. "A Lightweight Method for Generating Multi-Tier JIT Compilation Virtual Machine in a Meta-Tracing Compiler Framework (Artifact)." In: *Dagstuhl Artifacts Series* 11.2 (2025), 16:1–16:4. ISSN: 2509-8195. DOI: 10.4230/DARTS.11.2.16. URL: https://drops.dagstuhl.de/entities/document/10.4230/DARTS.11.2.16.
2. Yusuke Izawa, Hidehiko Masuhara, Carl Friedrich Bolz-Tereick, and Youyou Cong. "Threaded Code Generation with a Meta-Tracing JIT Compiler." In: *Journal of Object Technology* (2022), 2:1–11. ISSN: 1660-1769. DOI: 10.5381/jot.2022.21.2.a1. arXiv: 2106.12496.
3. Shusuke Takahashi, Yusuke Izawa, Hidehiko Masuhara, and Youyou Cong. "An approach to collect object graphs for data-structure live programming based on a language implementation framework." In: *Journal of Information Processing* 30 (2022), pp. 451–463. DOI: 10.2197/ipsjjip.30.451.

Conference Proceedings

1. Yusuke Izawa, Junichiro Kadomoto, and Hidetsugu Irie. "VisMorph: A Live Programming Environment for Shape-Adaptive Computers." In: *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. UIST Adjunct '25. Busan, Korea: Association for Computing Machinery, 2025. ISBN: 9798400720369. DOI: 10.1145/3746058.3758364. URL: https://doi.org/10.1145/3746058.3758364.

2 Yusuke Izawa, Hidehiko Masuhara, and Carl Friedrich Bolz-Tereick. "A Lightweight Method for Generating Multi-Tier JIT Compilation Virtual Machine in a Meta-Tracing Compiler Framework." In: *39th European Conference on Object-Oriented Programming (ECOOP 2025)*. Ed. by Jonathan Aldrich and Alexandra Silva. Vol. 333. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 16:1–16:29. ISBN: 978-3-95977-373-7. DOI: 10.4230/LIPIcs.ECOOP.2025.16. arXiv: http://arxiv.org/abs/2504.17460. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ECOOP.2025.16.

3 Yusuke Izawa, Junichiro Kadomoto, Hidetsugu Irie, and Shuichi Sakai. "Designing a Reactive Programming Language for Shape-Adaptive Computers." In: *2024 31st Asia-Pacific Software Engineering Conference (APSEC)*. 2024, pp. 452–456. DOI: 10.1109/APSEC65559.2024.00058.

4 Yusuke Izawa, Junichiro Kadomoto, Hidetsugu Irie, and Shuichi Sakai. "A Functional Reactive Programming Language for Wirelessly Connected Shape-Changeable Chiplet-Based Computers." In: SPLASH Companion 2023. Cascais, Portugal: Association for Computing Machinery, Aug. 30, 2023. ISBN: 979-8-4007-0384-3/23/10. DOI: 10.1145/3618305.3623608.

5 Yusuke Izawa, Hidehiko Masuhara, and Carl Friedrich Bolz-Tereick. "Two-level Just-in-Time Compilation with One Interpreter and One Engine." In: *The ACM SIGPLAN Workshop on Partial Evaluation and Program Manipulation*. PEPM 2022. Virtual, Jan. 17, 2022. arXiv: 2201.09268. URL: https://popl22.sigplan.org/details/pepm-2022-papers/3/Two-level-Just-in-Time-Compilation-with-One-Interpreter-and-One-Engine.

6 Yusuke Izawa and Hidehiko Masuhara. "Amalgamating Different JIT Compilations in a Meta-Tracing JIT Compiler Framework." In: *Proceedings of the 16th ACM SIGPLAN International Symposium on Dynamic Languages*. DLS 2020. Virtual, USA: Association for Computing Machinery, Nov. 17, 2020, pp. 1–15. ISBN: 9781450381758. DOI: 10.1145/3426422.3426977.

7 Hidehiko Masuhara, Shusuke Takahashi, Yusuke Izawa, and Youyou Cong. "Toward a Multi-Language and Multi-Environment Framework for Live Programming." In: *Proceedings of the 6th Workshop on Live Programming*. Live 2020. Virtual, 2020, pp. 1–5. URL: http://liveprog.org/live-2020/Toward-a-Multi-Language-and-Multi-Environment-Framework-for-Live-Programming/.

8 Yusuke Izawa, Hidehiko Masuhara, Tomoyuki Aotani, and Youyou Cong. "A Stack Hybridization for Meta-hybrid Just-in-time Compilation." In: *Proceedings of the 36th JSSST Annual Conference*. Ed. by Kei Ito. Japan Society for Software Science and Technology (JSSST). Shibaura Institute of Technology, Tokyo, Japan, Aug. 27, 2019, pp. 1–9. URL: http://jssst.or.jp/files/user/taikai/2019/proceedings.html.

9 Yusuke Izawa. "BacCaml: The Meta-Hybrid Just-in-Time Compiler." In: *Proceedings of the Conference Companion of the 3rd International Conference on Art, Science, and Engineering of Programming*. Programming 2019. Genova, Italy: Association for Computing Machinery, Apr. 2, 2019, pp. 1–3. ISBN: 9781450362573. DOI: 10.1145/3328433.3328466.

10 Yusuke Izawa, Hidehiko Masuhara, and Tomoyuki Aotani. "Extending a Meta-Tracing Compiler to Mix Method and Tracing Compilation." In: *Proceedings of the Conference Companion of the 3rd International Conference on Art, Science, and Engineering of Programming*. Programming 2019. Genova, Italy: Association for Computing Machinery, Apr. 2, 2019, pp. 1–3. ISBN: 9781450362573. DOI: 10.1145/3328433.3328439.

Open-Source and Professional Software

- Contributor to **PyPy/RPython**: meta-tracing infrastructure, multi-tier JIT generation, and tooling
- **nbjit.jl**: Julia-based experimental selective JIT compiler for notebook workflows
- **RPyForth**: Forth interpreter implementaton in RPython with tracing JIT support
- Live programming environment for shape-adaptive compilers (VisMorph)

Teaching Experience (Lead and Co-Lead)

Experiment on Compiler Construction

- Target: BEng 2nd year
- Teaching a compiler construction by building a comiler from *While* language to WebAssembly in OCaml

Experiment on Compiler Construction

- Target: BEng 3nd year
- Teaching hardware proramming by using a FPGA board and Verilog

Research Seminar on Programming Language and Virtual Machine

- Target: BEng 3nd year
- Leading a seminar that discusses programming language implementation and virtual machine construction

Introduction to Programming in C (II) (Co-Lead)

- Target: BEng 2nd year
- Teaching programming in C

Excersixe on Scripting Languages (Co-Lead)

- Target: BEng 3rd year
- Teaching programming in Python and small interpreter construction

Supervision

2025 — Supervised one BEng thesis on meta-tracing compilation for Forth

2024 — Supervised one BEng thesis on Ogasawara-hira bird ecology monitoring application for environmental conservation in Ogasawara (got Electrical Engineering Women's Activity Encouragement Award)

2022 — Mentored one MSc thesis on live data-structure programming environment (got the 2022 IPSJ Computer Science Research Award for Young Scientists)

Skills

Languages — English (fluent), Japanese (native), German (basic)

Coding — OCaml (S), Scala (S), Python (S), C (S), Java (A), Ruby (A), Shell (A), R (B), SQL (C)

Tool — LLVM, PyPy/RPython, Truffle/GraalVM, FPGA toolchains, Arduino

Teaching — lecture capture, online learning platform, live coding, inclusive assesment (rubrics, anonymised markong)